Deconstructing Requirements into Scenarios

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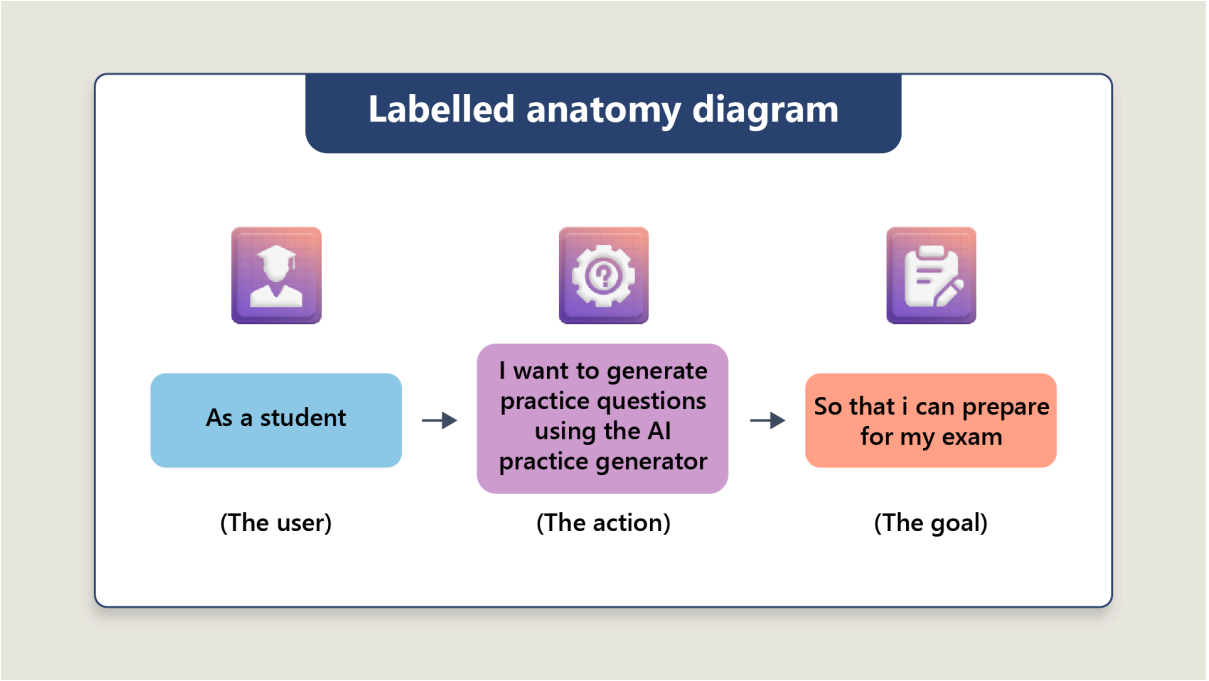
Introduction

A single software requirement, which often looks like a simple sentence, contains many different paths a user can take. To test a feature thoroughly, you must first break down that requirement into testable ideas called scenarios. This reading explains how to analyze a requirement, like a user story, to identify its different user journeys, recognize the concepts of Happy Paths and Negative Paths, and use business risk to prioritize which scenarios matter most.

Unpacking the User Story

In modern software development, requirements are often written as user stories: short, simple descriptions of a feature told from the perspective of the person who uses it. They follow a familiar template: As a [type of user], I want to [perform some action] so that I can [achieve some goal].

For example: As a student, I want to generate practice questions using the AI Practice Generator so that I can prepare for my exam.



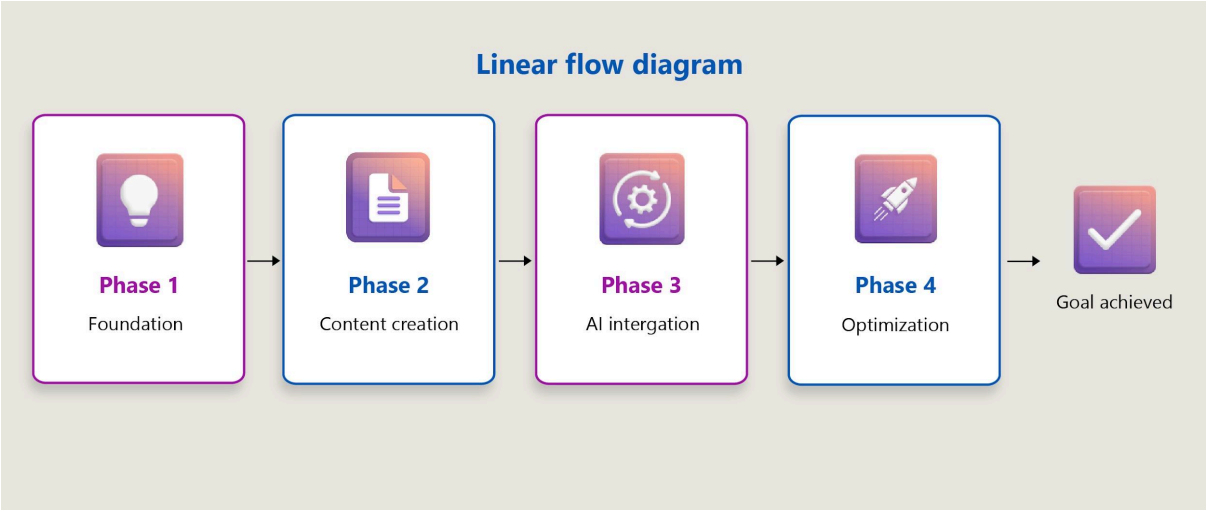
The image shows a section from a document titled Unpacking the User Story. It features a breakdown of an example user story sentence into three color-coded blocks to explain its structural anatomy. The sample sentence reads: As a student, I want to generate practice questions using the AI Practice Generator so that I can prepare for my exam. The first part of the sentence, "As a student," is highlighted in blue and labeled "The User." The second part, "I want to generate practice questions using the AI Practice Generator," is highlighted in green and labeled "The Action." The final part, "so that I can prepare for my exam," is highlighted in orange and labeled "The Goal."

This single sentence is your starting point. It describes the goal, but not all the ways a student might interact with the feature. Your job is to think through those possibilities.

The Happy Path: The Ideal Journey

The Happy Path is the ideal, straightforward journey where a user encounters no errors and successfully achieves their goal. For the AI Practice Generator, it looks like this:

1. The student is logged in with an active course enrollment.
2. The student enters a valid topic into the AI Practice Generator.
3. The generator processes the input and returns relevant practice questions.
4. The student reviews the questions and prepares for their exam.



The image presents a screenshot from a document detailing test path scenarios for an AI Practice Generator. It features a highlighted flowchart block that visually outlines a four-step linear sequence for the feature's Happy Path. The process flows horizontally from left to right through four green rounded rectangles connected by directional arrows. The first step indicates that the student is logged in with an active course enrollment. The second step shows the student entering a valid topic into the AI Practice Generator. In the third step, the generator processes that input to return relevant practice questions. The fourth and final step shows the student reviewing the questions to prepare for their exam. The entire sequence concludes on the far right with a distinct checkmark icon labeled "Goal Achieved". This diagram serves to make the ideal, error-free user journey concrete and easily understandable for software testing purposes.

Testing the Happy Path first is critical—if the core functionality does not work, nothing else matters.

Negative Paths: When Things Go Wrong

Negative Paths are scenarios where things do not go as planned due to user error, system issues, or unexpected conditions. These paths test the feature's resilience. For the AI Practice Generator, consider:

- What if the student submits an empty input field?
- What if the student enters an unsupported topic or special characters?
- What if the student is not logged in?
- What if the system times out?
- What if the student's course enrollment has expired?

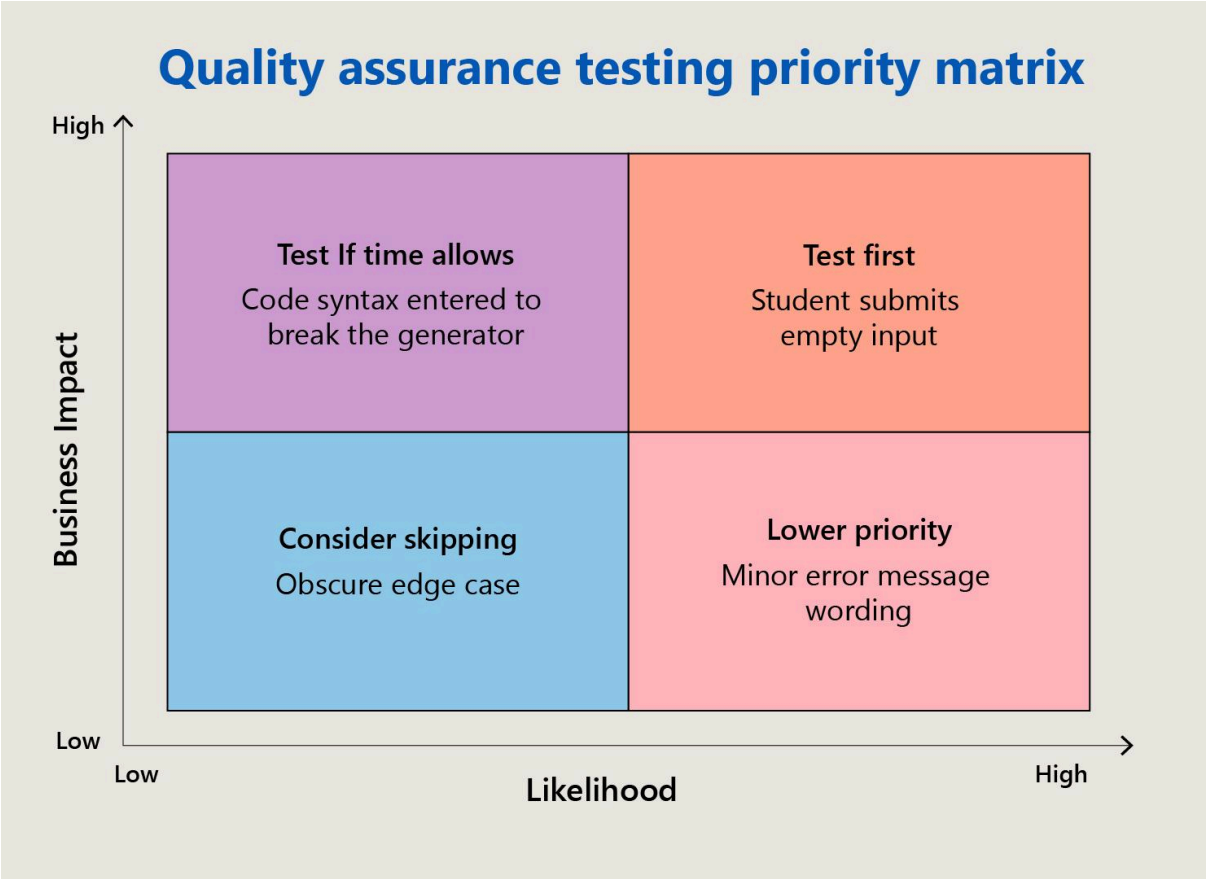
A well-built application handles these situations gracefully, with clear, helpful error messages rather than crashes or silence.

Prioritizing Scenarios Based on Risk

You cannot test every possible scenario, so you must prioritize based on risk, which is usually assessed using a combination of two main factors:

- Business Impact: How severe would the consequences be if this scenario failed?
- Likelihood: How often will a user actually encounter this scenario?

High-impact, high-likelihood scenarios, like a student submitting an empty input, should be tested first. Low-impact, low-likelihood scenarios can be deprioritized when time is limited.



The image features a highlighted section within a testing document showcasing a two-by-two Priority Matrix. The matrix maps testing scenarios based on two distinct axes: a horizontal axis representing likelihood from low to high, and a vertical axis

representing Business Impact from Low to High. The grid is divided into four distinct, color-coded quadrants to guide testing prioritization. The top-right, high-impact, high-likelihood quadrant is colored red and labeled "Test First", containing the scenario "Student submits empty input". The top-left, high-impact, low-likelihood quadrant is labeled "Test If Time Allows" and contains the scenario "Code syntax entered to break the generator". Moving to the lower tiers, the bottom-right, low-impact, high-likelihood quadrant is labeled "Lower Priority" and contains "Minor error message wording". Finally, the bottom-left quadrant is colored blue, labeled "Consider Skipping", and contains "Obscure edge case".

Conclusion

Deconstructing a requirement means seeing the many potential stories hidden within a single sentence. By identifying the Happy Path and then exploring the various Negative Paths, you build a comprehensive list of test scenarios. Prioritizing those scenarios by business risk ensures you protect users from the most critical and most common failures first.